



**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA
ARAS 7 & 8 BLOK D4, KOMPLEKS D
PUSAT PENTADBIRAN KERAJAAN PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

KOD DAN NAMA PROGRAM / <i>PROGRAM'S CODE AND NAME</i>	Z-009-1:2015CORE ABILITIES	
TAHAP / LEVEL	1 (ONE)	
NO. DAN TAJUK KEBOLEHAN TERAS / <i>CORE ABILITY NO. AND TITLE</i>	CA04 HEALTH, SAFETY AND ENVIRONMENTAL AWARENESS	
NO. DAN PENYATAAN KEBOLEHAN / <i>ABILITY NO. AND STATEMENT</i>	4.1 ADHERE TO HEALTH AWARENESS ACTIVITY 4.2 ADHERE TO SAFETY AWARENESS ACTIVITY 4.3 ADHERE TO ENVIRONMENTAL AWARENESS ACTIVITY	
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TAJUK/TITLE: SAFETY AND HEALTH WORKPLACE SYSTEM

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the Safety and Health Workplace System. The person who is competent in this CA shall be able to abide to Health, Safety and Environmental rules and regulations, hence reducing exposure to hazards and risks as well as incidence of work-related injury and illness.

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PENERANGAN/INFORMATION:

1.1 Type Of Hazard

Hazard means a source or a situation with a potential for harm in terms of human injury or ill health, damage to property, damage to the environment or a combination of these.

1.1 Hazard Identification

The purpose of hazard identification is to highlight the critical operations of tasks, that is, those tasks posing significant risks to the health and safety of employees as well as highlighting those hazards pertaining to certain equipment due to energy sources, working conditions or activities performed. Hazards can be divided into three main groups, health hazards, safety hazards, and environmental hazards.

i) Health hazards

An occupational health hazard is any agent that can cause illness to an individual. A health hazard may produce serious and immediate (acute) affects, or may cause long-term (chronic) problems. All or part of the body may be affected. Someone with an occupational illness may not recognize the symptoms immediately. For example, noise-induced hearing loss is often difficult for the affected individual to detect until it is well advanced.

ii) Safety hazards

A safety hazard is any force strong enough to cause injury, or damage to property. An injury caused by a safety hazard is usually obvious. For example, a worker may be badly cut. Safety hazards cause harm when workplace controls are not adequate. Some examples of safety hazards include, but are not limited to -

- a) slipping/tripping hazards (such as wires run across floors);
- b) fire hazards (from flammable materials);
- c) moving parts of machinery, tools and equipment (such as pinch and nip points);
- d) work at height (such as work done on scaffolds);
- e) ejection of material (such as from moulding);
- f) pressure systems (such as steam boilers and pipes);
- g) vehicles (such as forklifts and trucks);

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- h) lifting and other manual handling operations; and
- i) working alone.

1.2 Hazard identification technique

The employer shall develop a hazard identification and assessment methodology taking into account the following documents and information -

- i) any hazardous occurrence investigation reports;
- ii) first aid records and minor injury records;
- iii) work place health protection programs;
- iv) any results of work place inspections;
- v) any employee complaints and comments;
- vi) any government or employer reports, studies and tests concerning the health and safety of employees;
- vii) any reports made under the regulation of Occupational Safety and Health Act,1994
- viii) the record of hazardous substances; and
- ix) any other relevant information.

2. Work Instruction, Safe Operating Procedure And Emergency Response Plan

Making safe work procedures and practices part of standard operating procedure may seem a matter of common sense, but in fact an effective health and safety program for workers is required by the Occupational Health and Safety Regulation Without a written standard to refer to, whether for bed-making or respirator-wearing, there is no consistent guidance for workers or supervisors. Measurement of these items in the audit will include written safe work procedures, practices and/or instructions including:

- i) all routine and non-routine expected operations of the company
- ii) a Workplace Hazardous Materials Information System (WHMIS)
- iii) instructions that direct the first aid services, supplies and equipment to be provided and how employees receive that service
- iv) procedures addressing possible emergencies, training of workers in those procedures, testing their effectiveness, and evaluating and revising the procedures based on drills and actual emergencies.

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2.1 Work Instruction

A work instruction is a tool provided to help someone to do a job correctly. This simple statement implies that the purpose of the work instruction is quality and that the target user is the worker. Safe work instructions highlight the hazards and risks in a process or task and provide information necessary to assist all workers to perform activities safely. These instructions also assist in the training and orientation of new workers to reduce the risk of the tasks to be performed, as well as providing them with the rules and procedures necessary to ensure that they can perform their work in a safe manner.

2.2 Safe Operating Procedures (SOP)

A safe operating procedure is a written document that provides step-by-step instructions on how to safely perform a particular task in the workplace. Having a safe operating procedure in place will help train the employees to work safely, help identify risks and hazards and, importantly, help demonstrate to safety regulators that company is committed to improving safety in your workplace.

i) The Importance of Safe Operating Procedures

Safety operating procedures are an important internal control. Industries use safety operating procedures to ensure owners and employees do not injure themselves or customers during business operations. Certain industries may be subject to more safety operating procedures than others. Manufacturing or production, energy, construction and warehousing represent a few examples of industries using copious safety operating procedures. Employer can also use safety operating procedures to limit their company liability from dangerous or inappropriate actions. The importances of safe operating procedures are:

a. Decrease Incidents

Safety procedures can help decrease the number of workplace incidents. Many companies use an employee manual or guidebook to inform and educate employees on safety procedures. Operational managers are also an important part of enforcing safety procedures. Managers are responsible for reviewing employee actions and making sure company policies provide a safe workplace. Decreasing the number of employee safety incidents can also help companies save money on their insurance

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policies. Companies with several safety incidents are usually considered high risk by insurance companies.

b. Maintain Certifications

Employer may need to implement safety procedures to maintain third-party certifications. Third-party certifications usually provide companies with a seal of approval for their company operations. This approval can help companies save money on insurance policies or improve their goodwill with consumers. Organizational certification is usually specific to business industries. Small businesses operating in the energy industry may receive a different certification from safety procedures than a company operating in the retail business industry.

c. Legal Liability

Safety procedures can help companies reduce their legal liability. Legal liabilities often occur when employees or customers are injured during the company regular course of business. Injured employees may have less legal options than customers. Many employees sign arbitration agreements that allow the company to limit their loss from safety procedure incidents. Customers injured from lack of safety procedures can sue using a lawyer or petition a watchdog group to audit the company. Small businesses usually have fewer resources for protecting themselves against major legal liabilities. Safety procedures may help prevent this type of exposure in the business environment.

d. Improve Operations

Companies can improve business operations through the use of safety procedures. Improving operations may be an unintended benefit of safety procedures. Business owners and managers that educate employees on how to best complete business functions may find new ways to improve the efficiency and effectiveness of the production process. Safety procedures may also allow employees to work quicker and improve their production output. This can also help drive down business costs by using less employees to maintain specific levels of production output.

2.3 Emergency Response Plan (ERP)

Emergencies can create a variety of hazards for workers in the impacted area. Preparing before an emergency incident plays a vital role in ensuring that employers and workers have the necessary equipment, know where to go, and know how to keep them safe when an emergency occurs.

In Malaysia, the management of disaster is governed by the National Security Council (NSC) Directive No. 20: Policy and Mechanism of Disaster Management and Relief Committee. The Directive interpreted “Disaster” and “Non-disastrous incidents”. In addition, at the moment, the specific act and regulation which deal with the safety at the workplace is the “Occupational Safety and Health Act (OSHA) 1994”.

2.4 Emergency Response Plan (ERP) Procedures

The purpose of this procedure is to establish an emergency management procedure by:

- i) assigning responsibilities for Emergencies Response Plan (ERP);
- ii) providing the emergency procedures at all levels of ERP that required for specific local areas in each academic/administrative centre to prepare adequately the emergency procedures; and
- iii) managing the minimum evacuation practice requirements for each building.

2.5 Responsibilities, Accountabilities and Authority

A. Head of Local Management

- i) Nominate Chief of Local Incident Commander and their Deputies as appropriate and submit the names to the Emergency Response Committee for approval and record.
- ii) In consultation with Incident Commander, Chief of Local ERP Incident Commander and their Deputies must develop, establish and implement the local set up of ERP Procedure including Local Emergency Instructions.
- iii) In consultation with incident Commander, Chief of Local ERP Incident Commander and their Deputies must identify suitable Assembly Points for evacuations.

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- iv) All development and implementation of ERP Procedure must comply with the requirements and arrangements of objectives and related procedures.
- v) Ensure that ERT personnel have been provided with colour-coded helmets and/or vests.
- vi) All changes to ERT personnel are communicated to the incident Commander and Emergency Response Committee.
- vii) In consultation with Local ERT, training or drill is organized and conducted within their “Area Responsibility” for the implementation of the Local Emergency Instructions.
- viii) Relevant university community in their “Area Responsibility” shall participate in the emergency training.
- ix) Monitor and evaluate the implementation and performance of the Local Emergency Instructions, from Evacuation Drill Reports, and reporting yearly to the Incident Commander on the implementation, with suggestions for improvements.

B. Emergency Response Team (ERT) Personnel

They are responsible for:

- i) implementing a set of Local Emergency Instructions for each of the areas and activities they control, as appropriate;
- ii) assuming responsibility for their areas and community during emergencies;
- iii) keeping up-to-date Local Emergency Instructions in their area;
- iv) conducting at least one evacuation drill per building per year in their “Area Responsibility”, and documenting the details in accordance with the Evacuation Drill Report;
- v) If there is overlapping in “Area Responsibilities”, the choice of ERT Personnel should primarily be guided by personal characteristics and availability rather than by considerations related to organisational or reporting structure.

C. The Incident Commander

He is responsible for:

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- i) Submitting the nominated names to the Emergency Response Committee for approval;
- ii) Monitoring the implementation of this procedure and related documents in accordance with the Incident, Accident and Emergency Management;
- iii) Conducting training and advice for the implementation of this procedure within “Area Responsibility”.

2.6 Emergency Evacuation Procedure

A. Local Emergency Instructions must:

- i) address the risks of fire and serious injury;
- ii) identify any other emergency situations relevant to the area;
- iii) list the steps to be followed in each case, including the method to be used to initiate an evacuation of the building;
- iv) make special provision for the evacuation of people with disabilities, if applicable to the area; and

B. Emergency Floor Plans

At least one Emergency Floor Plan signage must be displayed on each floor of each building. The display shall show the:

- i) Name or code of the building;
- ii) Floor level;
- iii) Location of the plan itself (e.g. “You are here”) on the particular floor;
- iv) Emergency exits;
- v) Location of fire fighting equipment;
- vi) Location of any break-glass alarm;
- vii) Location of first aid kit;
- viii) Local Emergency Instructions displayed adjacent to the floor plan.

3. Personnel Protective Equipment (PPE)

Personal protective equipment, commonly referred to as "PPE", is equipment worn to minimize exposure to serious workplace injuries and illnesses. These injuries and illnesses may result from contact with chemical, radiological, physical, electrical, mechanical, or other workplace hazards. Personal protective equipment may include items such as gloves, safety glasses and shoes, earplugs or muffs, hard hats, respirators, or coveralls, vests and full body suits.



Picture 1: Personnel Protective Equipment

3.1 Proper Use Of Personal Protective Equipment (PPE)

All personal protective equipment should be of safe design and construction, and should be maintained in a clean and reliable fashion. It should fit well and be comfortable to wear, encouraging worker use. If the personal protective equipment does not fit properly, it can make the difference between being safely covered or dangerously exposed. When engineering, work practice, and administrative controls are not feasible or do not provide sufficient protection, employers must provide personal protective equipment to their workers and ensure its proper use. Employers are also required to train each worker required to use personal protective equipment to know:

- i) When it is necessary
- ii) What kind is necessary
- iii) How to properly put it on, adjust, wear and take it off
- iv) The limitations of the equipment
- v) Proper care, maintenance, useful life, and disposal of the equipment

If PPE is to be used, a PPE program should be implemented. This program should address the hazards present; the selection, maintenance, and use of PPE; the training of employees; and monitoring of the program to ensure its ongoing effectiveness. Personal protective equipment is addressed in Factories and Machinery Acts and Regulations, General Industry, Shipyard Employment, Marine Terminals, and Long shoring.

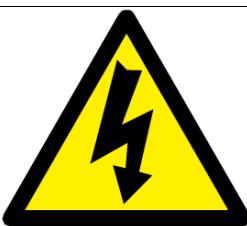
4. Safety And Health Signage

Safety signs must be used whenever a hazard or danger cannot be avoided adequately or reduced in another way. Before installing safety signs an employer should examine whether the hazard can be avoided or reduced by collective precautions (precautions that protect everybody) or safer ways of doing the work.

A safety sign provides information about safety or health and can be a signboard, colour, acoustic signal, verbal communication, or hand signal. A signboard is a sign that provides information or instruction using a combination of shape, colour and symbols but excludes information in writing. Employers must provide information to employees on the meaning and requirements of any signs used in the workplace, especially where text on supplementary signboards is used.

4.1 Type Of Safety Signboard

Safety signboard is a sign which provides information or instructions by a combination of shape, colour and a symbol or pictogram which is rendered visible by lighting of sufficient intensity. In practice many signboards may be accompanied by supplementary text (e.g. 'Fire exit' alongside the symbol of a moving person. Signboards can be of the following types:

Description	Signboard
a) prohibition sign - a sign prohibiting behaviour likely to increase or cause danger (e.g. 'no access for unauthorized persons');	
b) warning sign - a sign giving warning of a hazard or danger (e.g. 'danger: electricity')	
c) mandatory sign – a sign prescribing specific behaviour (e.g. 'eye protection must be worn');	

d) emergency escape or first-aid sign – a sign giving information on emergency exits, first aid, or rescue facilities (e.g. 'emergency exit/escape route').



Table 1: Safety Signboard

4.2 Colours And Shapes On Safety Signboards

Safety signs and colour are useful tools to help protect the health and safety of employees and workplace visitors.

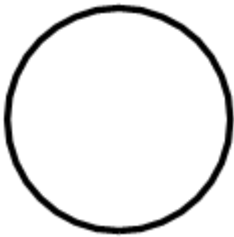
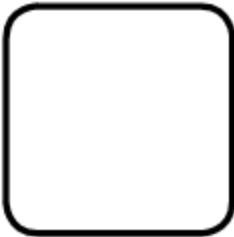
Category	Sub-category	Colour
 1. Regulatory A circle indicates that an order is in force	1.1 Prohibition forbids an action 1.2 Mandatory requires an action	Red and black on white White on black
 2. Warning A triangle indicates caution or danger	2.1 Caution indicates a potential hazard 2.2 Danger indicates a definite hazard	Black on yellow White on red
 3. Information A square indicates information	3.1 Emergency indicates first aid, health, fire protection, fire fighting and emergency equipment 3.2 General information indicates permission or public information	White on green White on blue

Table 2: Sign Categories

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SOALAN/QUESTION :

1. What is the purpose of hazard identification?

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2. What is the purpose of emergency response plan?

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3. What is the responsible of employers before providing the PPE to their employee?

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RUJUKAN / REFERENCES:

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2. Occupational Safety and Health Act 1994 and Regulation, (2009) Malaysia, Penerbitan Akta (M), ISBN-967701188-X.
3. Garis Panduan bagi Pengenalpastian Hazard, Penaksiran Risiko dan Kawalan Risiko (HIRARC), ISBN 978-983-2014-62-1